

# Quick Guide on Elle

## Intro

This doc is to provide a simple, quick guide on how to run Elle on a cluster using a container. A large portion of this tutorial is based on the [Bons book](#), and Steinbach's [PhD thesis](#). The example in this tutorial runs on Sherlock, the Stanford cluster and RCC, the UChicago cluster .

## Environment setup

The program will run inside a container. The only thing you need to have is the singularity container on your machine. To check whether you have it, type

None

```
singularity --version
```

## Quick overview on Elle

Before you start running anything, it is important to understand what Elle is, what types of files it needs to run, and how to visualize the outputs.

Elle initially runs on Ubuntu 14.04 (very old). It is a very very graphically intensive program. So to save all the troubles of getting the correct operating system, package installations (and version compatibility), I built Elle with ice in it into a container that you can just directly launch.

Elle is a 2D solver that takes in an 2D image of grains, along with other input files, such as phase info, and runs through a batch script.

The core file of the program is .elle. It contains: the graphical information of the shapes and locations of the grains, euler angles, temperature, pressure, time step which the simulation is performed on, and many other

things. This file basically tells you what your microstructure domain looks like. I summarized some of the key features of .elle in the next section, but to see more detailed and structured info on .elle file, please refer to *Appendix J The Elle file format* in Bons book.

Other input files include: `phase_db.txt`, `air.sx`, `ice3d.sx`, `symmetry.symm`. These are the files specifying the physical properties of ice structure. You typically don't need to change any of these. `ppc.in` specifies the input velocity gradient tensor that will be exerted on the microstructure.

Lastly, you may want to change the launch file `launch.sh` according to your needs. This batch script has a series of commands that run the program.

## .elle file

This is the heart file for the Elle program. It specifies the microstructure shapes, size, euler angle, and many other properties. When you run the program, you need an .elle as input microstructure. As the program runs, it will output an .elle at each time step, representing the microstructure at that time step. .elle has three categories:

- 1) Environmental condition - top several lines in the file. You typically only need to change the temperature and time step here.
- 2) "Digitized" microstructure properties, including their shapes, euler angles.
- 3) Structured grid. Currently Elle can only handle 128x128 or 256x256.

There are two sets of grids in .elle, unodes and bnodes. The grains are represented by polygons - by connecting the nodes to form one grain. These nodes are referred to as bnodes. Unodes is the structured grid. See Steinbach's dissertation for more information.

The FLYNN section in .elle is where you specify the shapes. Each row under the FLYNN section represents one grain. For example, if you have a rectangular shaped grain, and the IDs for the four bnodes are 11, 12, 13, 14. Then the first row is

None

```
0 11 12 13 14
```

where the first 0 is the ID of the FLYNN, following numbers are the IDs for the bnodes that form this grain. So connecting 11 -> 12 -> 13 -> 14 will form FLYNN 0.

Note that the 11, 12, 13, 14 are the IDs of the bnodes, not the spatial coordinates of those points. The spatial coordinates are stored in the LOCATION section, e.g.

None

```
LOCATION
```

```
0 0.0050000000 0.9900000000
```

```
1 0.0100000000 0.9850000000
```

```
2 0.1550000000 0.9900000000
```

```
...
```

where the first number in each row is the ID of the bnodes, and following are the x and y coordinates.

## Open container using singularity

There are many ways to open the container depending on what you want to do. Before opening the container, we need unset the mount flag, so that the cluster's directory does not overwrite the container's directory which has all the executables.

None

```
unset SINGULARITY_BINDPATH APPTAINER_BINDPATH \  
SINGULARITY_HOME APPTAINER_HOME \  
SINGULARITYENV_HOME APPTAINERENV_HOME
```

If you only want to run some program on the container without changing any source codes (which is probably most common), run

None

```
singularity shell --no-home --cleanenv -B /project -B /scratch -B /software -B  
"$HOME":/host_home elle_ice.sif
```

The `-B /project..` flag allows you to mount the clusters directory (which will contain your local input/output files etc.) to your container. This command will open the container. Then set the paths using

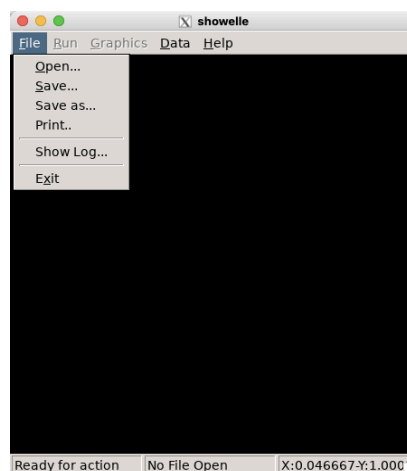
None

```
export PATH=$PATH:/home/emmalieu/programs/elle/elle/binwx  
export LD_LIBRARY_PATH=/lib:/usr/lib:/usr/local/lib  
export ELLEPATH=/home/emmalieu/programs/elle/elle/binwx  
export PATH=${PATH}:${ELLEPATH}:
```

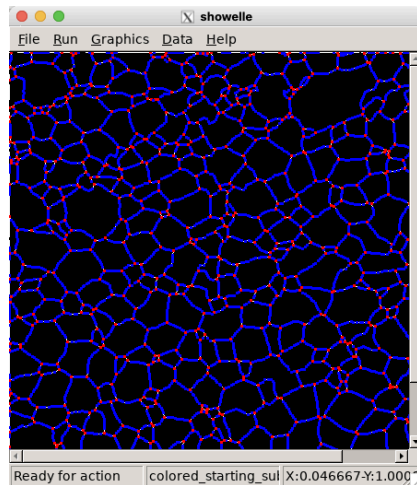
Now you can call Elle commands there.

## Visualize microstructure

There are two ways to visualize a microstructure. First, after you open a container (previous section), type `showelle`. This will prompt an old looking GUI like the following



Then you can open your .elle file under File -> Open. For example, some microstructure may look like this



## Run Elle simulations

I provided a small example of a simple shearing experiment under the example folder. In this experiment, the microstructure (fine\_foam.elle) is being sheared at a rate of  $10^{-9}$  1/s, which is specified in `ppc.in`. To run the script, simply type

None

```
./launch.sh
```

after you enter the container. The simulation runs only 10 steps. The results are saved in the results folder, under which you can find the .elle for each time step.

If you want to experiment with your own settings, e.g. pure shear or more complex condition, different temperature, different time step, different input microstructure, make a copy of the entire folder and modify from there. It is not recommended to directly modify the example itself.